

Event & Room Occupancy Management System

Project Review 1 — Abstract & Design Proposal

TEAM

Sec 10 · Team 1

TRIMESTER

2026–2027, Tri-4

MEMBERS

J. Geya Reddy · Kamtam Priyanka

Problem Statement

Students and administrators lack a single, reliable way to see which campus rooms are free and to manage events without last-minute chaos.



Room availability is invisible

Students struggle to find free rooms during college hours — there is no shared, real-time view.



Spot registrations disrupt planning

Hackathons, workshops and seminars often see sudden participant surges beyond initial estimates.



Conflicts go undetected

Manual coordination leads to double-booked rooms and scheduling clashes.



Space is used inefficiently

Without occupancy tracking, capacity mismatches and overcrowding are common.

Existing System

How rooms & events are managed today

- Room status tracked informally — notice boards, spreadsheets, or word of mouth
- Event coordination happens over calls, messages and emails between organizers and admin staff
- No shared calendar view of bookings across departments
- Attendance and occupancy are estimated only after an event, if at all

LIMITATIONS



No single source of truth

Every stakeholder checks a different, often outdated, source.



Reactive, not proactive

Conflicts and overcrowding are discovered only after they occur.



Slow decision-making

Admins cannot quickly compare expected vs actual demand.



Not scalable

Manual coordination breaks down as the number of events grows.

Proposed System

A centralized, web-based platform where students discover rooms and events, and administrators allocate space, monitor registrations, and track real occupancy — all from one dashboard.



Live Room Availability

View real-time room status, capacity and schedules before booking.



Event Scheduling

Admins create and manage events, allocate rooms and set expected turnout.



Participant Registration

Students register for events; the system tracks registrations against capacity.



Conflict Detection

Automatically flags overlapping bookings and capacity mismatches.



Admin Dashboard

A single console to monitor bookings, rooms and occupancy trends.



Occupancy Tracking

Compares expected vs. actual attendance to guide future planning.

Objectives

1 Provide a centralized platform for room and event management across campus

2 Give students real-time visibility into room availability and schedules

3 Streamline event creation, room allocation and participant registration

4 Detect and prevent room-booking conflicts before they occur

5 Track expected vs. actual occupancy to support data-driven decisions

6 Improve overall utilization of campus spaces and reduce manual effort

Scope

IN SCOPE

- ✓ Room catalog with capacity, building & equipment details
- ✓ Event creation, scheduling and room allocation by admins
- ✓ Student registration for events with live seat tracking
- ✓ Booking conflict detection across rooms and time slots
- ✓ Occupancy dashboard comparing expected vs. actual turnout
- ✓ Role-based access for students and administrators

OUT OF SCOPE (for now)

- Online payment or fee collection for events
- Native mobile applications (web is responsive instead)
- Automated push notifications / SMS alerts
- Integration with external college ERP systems
- Advanced analytics or predictive forecasting
- Multi-campus / multi-institution support

Technology Identification



React.js

Frontend framework for the web interface



Node.js

Backend JavaScript runtime environment



Express.js

Backend framework for REST APIs



MySQL

Relational database for system data



MySQL Workbench

Database design & SQL development



Postman

Testing and debugging REST APIs



Visual Studio Code

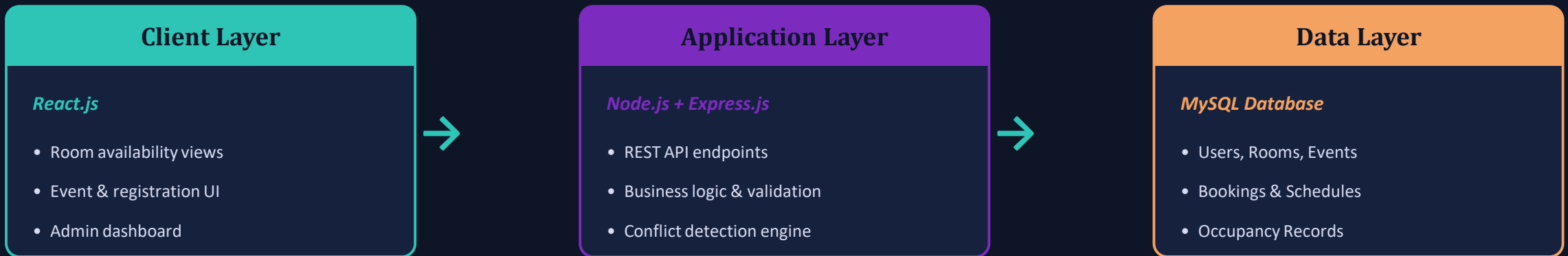
Primary development environment



Git & GitHub

Version control & source management

Basic Architecture



Client requests flow through the Express REST API, which applies business rules (e.g. conflict checks) before reading or writing to MySQL — keeping presentation, logic and data cleanly separated.

Team Details

TEAM 1 · SECTION 10



J. Geya Reddy

Roll No. 2520090123



Kamtam Priyanka

Roll No. 2520090001

GUIDE

Divya Priya Degala

COURSE CONTEXT

COURSE

25CS1302E — Database Systems Engineering & Distributed Backend Development

TERM

Trimester 4, 2026–2027

PROJECT

Event & Room Occupancy Management System

STACK

React · Node.js · Express.js · MySQL

Abstract Summary

The Event & Room Occupancy Management System replaces informal, manual coordination with a single web platform. Students get a clear view of room availability and can register for events confidently; administrators gain the tools to allocate rooms, resolve conflicts early, and track how spaces are actually used — turning guesswork into data-backed decisions for campus resource planning.

3-Tier

Architecture

6+

Core Entities

3NF

Normalized Schema

2

User Roles

Abstract submitted · Ready for Review 1 evaluation